

Grounding Language-Based Video Planning in Predictive Video Latents

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Abstract

High-level video planners that operate mainly in text space can generate step-by-step trajectories whose state changes sound correct (e.g., “water boiled”, “tea ready”) yet are not visually consistent with how the world evolves in real videos, a mismatch that worsens under domain shift (camera style changes, egocentric views) and noisy step annotations [1]. Existing Vision-Language World Model (VLWM) style systems compress video into language (actions and world-state changes) and select among candidate plans using a learned text critic, improving procedural coherence but still relying primarily on text-level self-consistency [1]. In parallel, V-JEPA-2 learns strong predictive video latents that support planning with visual goal images, but it does not directly ground language goals or textual intermediate state changes in its latent dynamics [2]. We propose to bridge these paradigms by grounding language planning in a predictive latent space. First, *state-change grounding* trains each predicted textual state-change span to match the corresponding V-JEPA’s latent transition between adjacent video segments. Second, *goal grounding* learns a lightweight encoder mapping goal text to a goal latent, enabling an energy objective that prefers plans whose predicted final latent is close to the goal latent. At inference, we generate K candidate plans and score each with a weighted sum of text critic cost, transition inconsistency penalty, and goal latent distance. We evaluate on COIN and CrossTask VPA benchmarks, WorldPrediction-PP, and a controlled retrieval-style consistency benchmark built from video segment pairs, measuring task success, robustness to paraphrase/domain shifts, compute overhead, and text–video consistency [4, 5, 6].

References

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